

Project 5 — License Plate Deblurring

Introduction to Computer Vision — Final Project
Blind Wiener Filter / FFT Deconvolution / CNN PSF Estimation

1. Introduction

Motion blur is a common degradation in traffic surveillance. When a vehicle moves during camera exposure, license plate pixels are smeared in the direction of motion, making text unreadable. This project implements a complete blind motion deblurring pipeline — at inference time, no ground truth PSF is provided. A CNN estimates blur parameters from the FFT magnitude, and the Wiener filter uses this PSF for frequency-domain deconvolution. The CNN is used solely for PSF estimation, in compliance with project guidelines. The core contribution is the Wiener filter.

2. Dataset

We use the Kaggle Car Plate Detection dataset — 433 real car photos with Pascal VOC XML annotations. Each plate is cropped and resized to 224x64 pixels, yielding 471 crops. The dataset is split 80/20: 376 training images and 95 validation images (random_state=42). Synthetic motion blur is applied with angle=30 degrees and length=25px. Gaussian noise (std=0.01) simulates real camera sensor noise.

The degradation model is:

$$g(x, y) = f(x, y) * h(x, y) + n(x, y)$$

where f is the original image, h is the PSF, n is Gaussian noise, and g is the observed blurred noisy image.

3. Approach

3.1 PSF Construction (Section 3)

The PSF is a normalized line segment of given length rotated to a given angle, built using `cv2.line` and `cv2.getRotationMatrix2D`. Normalization ensures total image brightness is preserved after convolution. Blur is applied in the frequency domain via the convolution theorem — $O(n \log n)$ vs $O(n^2)$ for spatial convolution. The PSF is aligned to the FFT origin using `np.roll` to prevent circular shift artifacts.

3.2 FFT Magnitude Analysis (Section 4)

Motion blur creates dark stripes in the FFT magnitude spectrum perpendicular to the blur direction. After suppressing the DC component, a radial scan across 180 directions finds the minimum energy direction. Correcting by 90 degrees recovers the estimated blur angle. This demonstrates the frequency domain analysis required by the project brief.

3.3 CNN-based PSF Estimation (Section 5)

A small CNN learns to predict blur angle and length directly from the FFT magnitude of the blurred image, making the pipeline fully blind. The architecture uses 4 convolutional layers with BatchNorm and ReLU, AdaptiveAvgPool, and two fully connected layers. The network outputs $[\cos(2\theta), \sin(2\theta), \text{normalized_length}]$ — encoding angle as cos/sin eliminates the 0=180 degree ambiguity inherent in motion blur. Training uses 4000 samples with random blur parameters (angle 0-180 degrees, length 5-40px) applied to real plate crops, 25 epochs, Adam optimizer with step LR decay.

3.4 Wiener Filter (Section 6)

The Wiener filter is the optimal linear deblurring filter under Gaussian noise, minimizing mean squared error between estimated and true image. Derived from minimizing $E[|F_{\text{hat}} - F|^2]$, the frequency-domain formula is:

$$F_{\text{hat}} = (H^* / (|H|^2 + K)) * G$$

G is the FFT of the blurred noisy image, H is the FFT of the PSF, H^* is the complex conjugate, K is the regularization parameter (noise-to-signal ratio), and F_{hat} is the estimated original image. K prevents division by near-zero at frequencies where the PSF response is weak — without K the naive inverse filter amplifies noise massively, scoring only 7.17 dB in our comparison.

3.5 Pre-processing

Before Wiener filtering, the blurred noisy image undergoes light Non-Local Means denoising (cv2.fastNlMeansDenoising, h=3). This gently reduces sensor noise while preserving edges, giving the Wiener filter a cleaner input.

4. Results

4.1 K Parameter Sweep (Section 7)

Six K values were tested to find optimal regularization:

K value	PSNR (dB)	Observation
0.0001	~21.9	Noisy — over-sharpened
0.001	~22.6	Improving
0.005	~22.8	Near optimal
0.010	~22.8	Optimal
0.020	~22.7	Slight over-smoothing
0.050	~22.1	Over-smoothed

Table 1: K parameter sweep results. Optimal K = 0.01.

4.2 Validation Set Evaluation (Sections 8-9)

The pipeline is evaluated on 95 validation images under two conditions: ground truth PSF (upper bound) and CNN-estimated PSF (blind). Results include mean and standard deviation:

Method	PSNR (dB)	SSIM	PSNR Gain
Blurred + Noisy	17.77 +/- 3.70	0.402 +/- 0.157	--
Wiener -- CNN PSF (blind)	21.67 +/- 3.88	0.641 +/- 0.121	+3.90
Wiener -- True PSF (upper bound)	23.31 +/- 3.84	0.727 +/- 0.069	+5.54

Table 2: Validation set results (95 images) with standard deviation.

The +3.90 dB blind improvement means MSE is reduced by a factor of 2.5. The 1.64 dB gap between CNN and true PSF represents the cost of blind estimation. Notably, SSIM standard deviation drops from 0.157 (blurred) to 0.121 (CNN restored), meaning restoration produces more consistent results across the dataset.

4.3 Four-Method Comparison (Section 11)

Four methods are compared to isolate each component's contribution:

Method	Avg PSNR	Noise Model	What it isolates
Naive Inverse Filter	7.17 dB	None	Baseline — no regularization
Lucy-Richardson	20.91 dB	Poisson	Wrong noise assumption
Wiener -- CNN PSF (blind)	23.26 dB	Gaussian	Our blind method
Wiener -- True PSF (upper bound)	25.34 dB	Gaussian	Perfect estimation

Table 3: Method comparison on 5 validation images.

Three key findings emerge. First, the naive inverse filter scores 7.17 dB — worse than the blurred input — proving K regularization contributes +16.09 dB and is essential. Second, Wiener outperforms Lucy-Richardson by +2.35 dB because Lucy-Richardson assumes Poisson noise while the actual noise is Gaussian — matching the noise model to the data matters. Third, the CNN closes most of the gap to perfect estimation — only 1.64 dB away from knowing the true PSF.

5. Challenges

The main technical challenge was PSF alignment in FFT convolution. Incorrect alignment using `ifftshift` instead of `np.roll` caused ghosting artifacts — the image appeared doubled. The fix was using `np.roll` to shift the PSF center to position (0,0), which is what FFT expects for circular convolution.

A second challenge was the 0=180 degree angle ambiguity — a horizontal blur looks identical regardless of direction. We resolved this by encoding angle as $\cos(2 \cdot \theta)$ and $\sin(2 \cdot \theta)$ in the CNN output, making both equivalent angles map to the same representation.

Finding optimal K required empirical sweep. The noise variance ($0.01^2 = 0.0001$) provided a theoretical starting point, confirmed by sweep results at $K=0.01$.

6. Limitations

The main limitation is the synthetic and fixed blur model — angle=30 degrees and length=25px applied identically to all validation images. Real motion blur varies per image and may include camera shake or spatially varying components not captured by a linear uniform PSF. The CNN also trains and evaluates on the same synthetic blur family, limiting generalization claims.

Future work would include testing on real motion-blurred traffic camera footage, random blur parameters per validation image, and a larger more diverse dataset to improve CNN generalization and close the 1.64 dB blind estimation gap.

7. Connection to Course Concepts

2D Convolution: Motion blur modeled as convolution with PSF. Convolution theorem enables efficient FFT-based implementation.

Fourier Transform: FFT used for fast blur application, frequency-domain blur signature analysis, and Wiener deconvolution.

Image Restoration: Wiener filter directly implements the minimum MSE estimator from deconvolution lectures.

Signal-to-Noise Ratio / PSNR: K = noise/signal ratio. PSNR measures restoration quality in decibels — every +3 dB halves MSE.

Structural Similarity (SSIM): Extends evaluation beyond pixel error to perceptual quality — luminance, contrast, and structure.

Regularization: K demonstrates the bias-variance tradeoff: small K amplifies noise, large K over-smooths.

8. Conclusion

We implemented a complete blind license plate deblurring pipeline achieving +3.90 dB PSNR and +0.239 SSIM across 95 real validation images using CNN-estimated PSF. The four-method comparison demonstrates: K regularization contributes +16.09 dB, correct Gaussian noise model contributes +2.35 dB over Lucy-Richardson, and the CNN achieves near-optimal blind estimation (1.64 dB gap to perfect). The pipeline is accompanied by an interactive Jupyter widget and browser-based HTML demo with live parameter sliders.

References

- [1] R. C. Gonzalez and R. E. Woods, Digital Image Processing, 4th ed. Pearson, 2018.
- [2] W. H. Richardson, 'Bayesian-Based Iterative Method of Image Restoration,' J. Opt. Soc. Am., vol. 62, no. 1, pp. 55-59, 1972.
- [3] Z. Wang, A. C. Bovik, H. R. Sheikh, E. P. Simoncelli, 'Image Quality Assessment: From Error Visibility to Structural Similarity,' IEEE Trans. Image Process., vol. 13, no. 4, 2004.
- [4] Kaggle Car Plate Detection Dataset. <https://www.kaggle.com/datasets/andrewmvd/car-plate-detection>